English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

Probability and Statistics, CS Year 3, 2020/2021 Test 2 *Your grade is 2 + the number of points. Time allowed: 3 hours. Good luck! Work on* $(\Omega, \mathcal{F}, \mathbb{P})$.

Problem 1. A tank is filled with $X \in [0,1]$ units at the start of the week, and Y units are consumed during the week. Their joint density is

$$f_{X,Y}(x,y) = 3x\mathbb{1}_{\{0\le y\le x\le 1\}} = \begin{cases} 3x, & 0 \le y \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

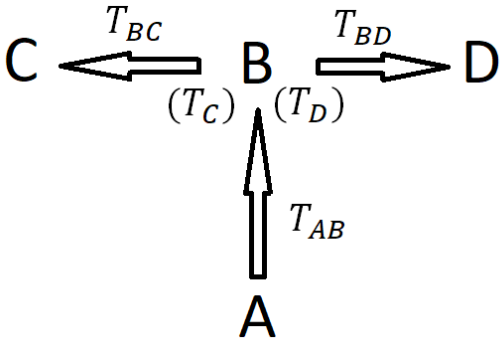
1. (0.25 points) Find the probability that at most $1/2$ unit is added and more than $1/4$ consumed.
2. (0.75 points) Find $\mathrm{Cov}(X,Y)$.

Problem 2. (1 point) Information displays at public transport stops contain 10×100 diodes. Their high-quality components have exponentially distributed working lifetimes with mean 10 years. When fewer than 75% work, the information is often unreadable and repair is required. What is the probability of repair being needed after three years?

Problem 3. Model returns of assets A, B, C as independent $N(3,2)$, $N(3,3)$ and $N(1,10)$ variables. You have five investment units.

1. (0.25 points) How should you allocate them to maximize expected profit?
2. (0.25 points) Among those allocations, which minimizes variance?
3. (0.5 points) A risk-taking investor puts five units into independent asset $D \sim N(-2,20)$. What is the probability this outperforms part 2?

Problem 4. Travellers X, Y go together from A to B, then wait for buses to C, D, respectively. The travel times are $T_{AB} \sim \mathrm{Exp}(3)$, $T_{BC} \sim \mathrm{Exp}(4)$ and $T_{BD} \sim \mathrm{Exp}(5)$; the waiting times at B are $T_C \sim \mathrm{Exp}(1)$ and $T_D \sim \mathrm{Exp}(2)$. All these times are independent. Let ξ, η be the total journey times.



1. (0.25 points) Find $\mathbb{P}(T_C + \ln(\mathbb{E}T_D) > 0)$.
2. (0.75 points) Find $\mathrm{Cor}(\xi, \eta)$.